GGG-C-735B
February 29, 1988
SUPERSEDING
GGG-C-735A
August 4, 1965

FEDERAL SPECIFICATION

CUTTER, ANGLE, HAND OPERATED

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification establishes requirements for hand operated angle cutters used to cut angle iron and slotted angle material.

1.2 CLASSIFICATION

1.2.1 Types and classes. Angle cutters shall be of one type and of the following classes as specified (see 6.1).

Class 1 - Right angle feed

Class 2 - End feed (guillotine)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools

Federal Standard:

Federal Standard H-28 - Screw-Thread for Federal Services

(Activities outside the Federal Government may obtain copies of Federal specification, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

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FSC 5110

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

ASTM E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The material shall be as specified herein.

3.2 Finish. Cutters shall have smooth, even surfaces and shall be thoroughly cleaned and free from scale, pits, nodules, burrs, cracks, or other detrimental defects.

3.3 Coating. All surfaces shall have a protective coating of enamel, lacquer or paint to prevent rusting in accordance with accepted industry practice.

3.4 Identification marking. Each cutter shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number, and country of origin.

3.5 Replacement parts. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification (see 6.1.1).

3.6 Screw threads. The screw threads shall be in accordance with Fed Std. H-28.

3.7 Hardness. Both the upper and lower cutting jaws shall be of steel and shall show hardness on the Rockwell "C" scale of 55 minimum to 60 maximum when tested in accordance with ASTM E-18.

3.8 Construction and design. Cutters shall be constructed for cutting and aligning the various commercial brands of steel and aluminum slotted angle (framing) up to and including 1-1/2 inches by 3 inches, 12 gauge steel material. Cutters shall meet the test requirements specified in 4.4.2. Cutters shall consist of an upper handle and upper cutting jaw, a lower handle and lower cutting jaw, and a work guide.

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3.8.1 Upper handle and upper jaw. The upper handle shall be made of steel with one end firmly fixed to the linkage. The linkage shall be connected to the upper jaw so that the upper jaw opens and closes as the upper handle is raised and lowered.

3.8.2 Lower handle and lower jaw. One end of the lower handle shall be firmly fixed to the lower jaw. The lower handle shall be of steel designed to steady the cutter during the cutter operation.

3.8.3 Class 1, right angle feed. Class 1 cutters shall be designed with the work guide mounted at right angles to the longitudinal axis of the cutter. The work guide shall not be less than 6 inches long and shall have at least one locator pin and one or more threaded locator holes. The locator pin and locator holes shall be designed and spaced so that they will properly align and index the slotted angle material.

3.8.3.1 Class 1 cutters shall weigh not more than 20 pounds and shall conform to the dimensions of, and be similar to figure 1.

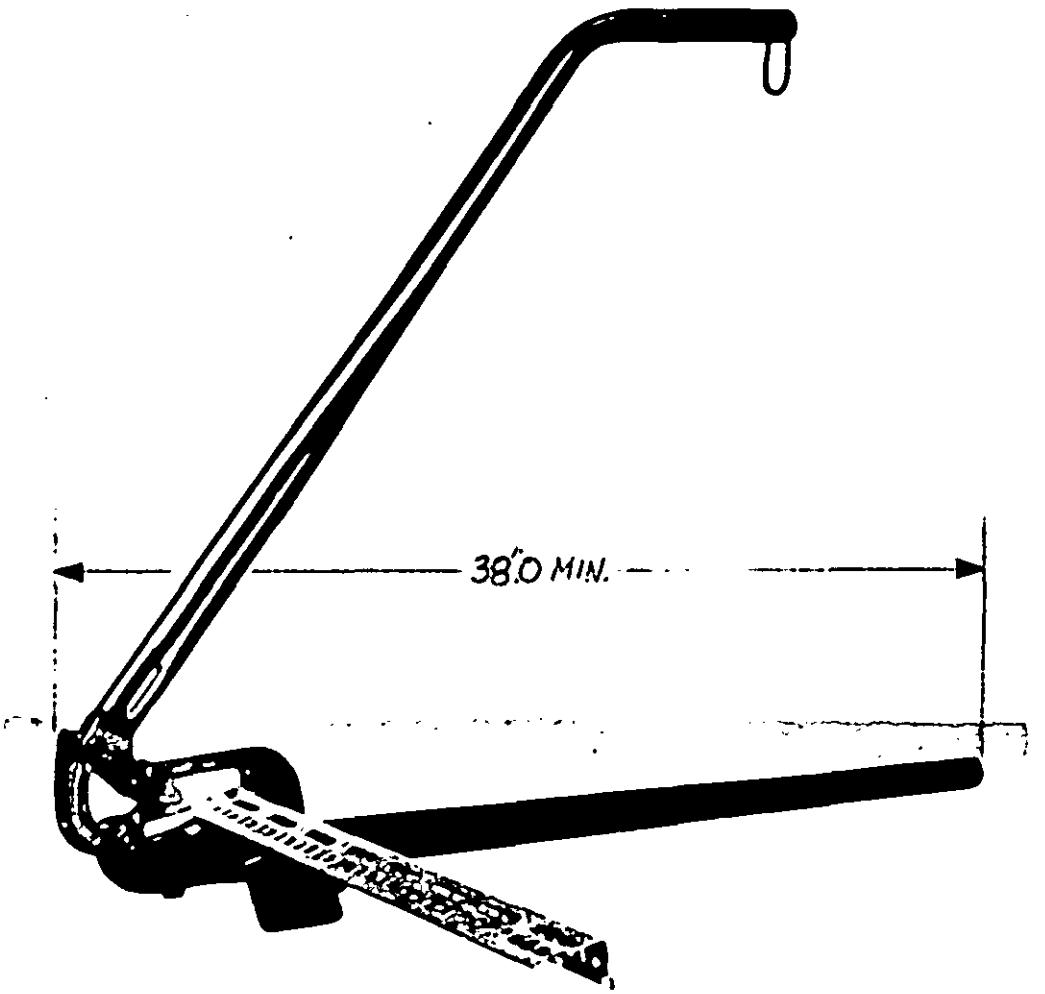


FIGURE 1. Class 1, right angle feed

3.8.4 Class 2, end feed (guillotine). Class 2 cutters shall be designed so that the work guide serves as the lower handle and is parallel to the longitudinal axis of the cutter. The work guide shall have at least one threaded locator hole and a locator pin (see A of figure 2). The locator pin and hole(s) shall be designed to align and index the slotted angle material.

3.8.4.1 Class 2 cutters shall weigh not more than 35 pounds and shall conform to the dimensions of, and be similar to figure 2.

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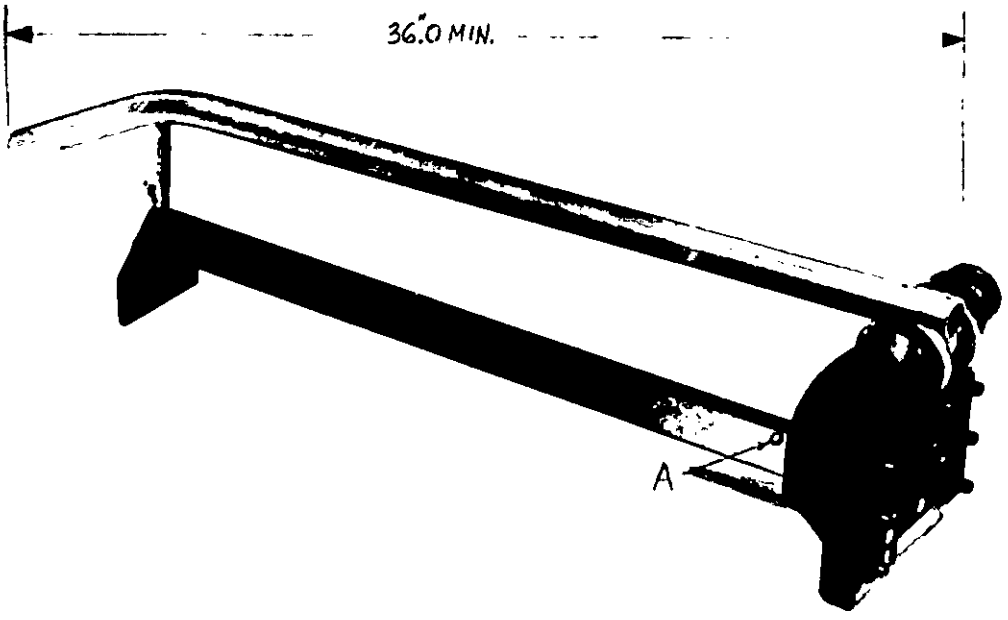


FIGURE 2. Class 2, end feed (guillotine)

3.9 Workmanship. The cutters shall be free from rust, burrs, fins, or any imperfections which may impair their appearance, durability or serviceability.

3.10 Regulatory requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and testing. Sampling shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes. The lot shall consist of all cutters offered for acceptance at one time. The sample unit is one cutter.

4.3 Inspection of the end item.

4.3.1 Examination for visual and dimensional defects. The inspection level shall be Level II and the AQL shall be 4.0, expressed in terms of percent defective. Any individual cutter in the sample lot containing one or more visual or dimensional defects shall be rejected.

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4.3.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5 and the contract. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.4 Testing. The cutters shall be tested in accordance with 4.4.1 and 4.4.2. The inspection level shall be S-3 with an AQL of 2.5 percent defective.

4.4.1 Hardness test. Each blade of the cutters shall be tested in accordance with ASTM E-18 to determine compliance with the requirements specified in 3.7. Hardness readings shall be made on each blade as close to the cutting edge as practicable.

4.4.2 Cutting test. The cutters shall be tested by cutting not less than 4 pieces of 1-1/2 by 3 inches 12 gauge slotted steel angle. This material shall be cut at 90° ± 2° and shall be cut on the arrow or indicating mark (on the material being cut). Any burrs found at the edges of the material cut shall not exceed .010 inch. After the test, all component parts of the cutter shall be examined and shall not show any evidence of fracture, distortion, or permanent set.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or Commercial, as specified (see 6.1).

5.1.1 Level A. Level A preservation and packaging shall be in accordance with level A of PPP-P-40.

5.1.2 Commercial. The cutters shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.1).

5.2.1 Level A. Level A packing shall be in accordance with level A of PPP-P-40.

5.2.2 Commercial. Cutters shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable to common carrier under the Uniform Freight Classification and the National Motor Freight Classification Rules.

5.3 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Ordering data. Purchasers should specify the following:

(a) Title, number, and date of this specification.

(b) Class of cutter (see 1.2.1).

(c) Applicable levels of preservation, packaging and packing (see 5.1 and and 5.2).

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6.1.1 When repair parts for cutters are required, the following should be specified:

(a) The manufacturer's name or trademark of cutter as required.

(b) The proper name of the part and part number.

(c) Quantity of each item to be procured.

(d) Any other data applicable to the cutter which will assist in securing the proper repair parts.

Military Coordinating Activity

Army - GL

Preparing Activity

GSA - FSS

Custodians:

Army - GL

Air Force - 99

Review Activity:

Air Force - 84

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